Chapter 5 : Exponents

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I - Exponents

a - Fundamental laws of exponents

Investigation

When a packet of peas is placed in the freezer, its temperature after t minutes is given by

$$T(t) = 0 + 32 \times 2^{-0.2t} \ ^\circ\text{C}.$$

Find the temperature of the peas:

1. initially
2. after 5 minutes
3. after 15 minutes

What does the graph of this function look like?

Property

The exponent laws for $a, b \in \mathbb{R} \setminus \{0\}$ and $m, n \in \mathbb{Z}$ are:

$$a^m \times a^n = a^{m+n}$$
$$\frac{a^m}{a^n} = a^{m-n}$$
$$(a^m)^n = a^{mn}$$
$$(ab)^n = a^n b^n$$
$$\left(\frac{a}{b}\right)^n = \frac{a^n}{b^n}$$
$$a^0 = 1$$
$$a^{-n} = \frac{1}{a^n}, \quad \text{and in particular } a^{-1} = \frac{1}{a}$$

Exercise 1. Write as a single power of 2:

1. 2×2^a
2. 4×2^b
3. 8×2^t
4. $(2^{x+1})^2$
5. $(2^{1-n})^{-1}$
6. $\frac{2^c}{4}$
7. $\frac{2^m}{2^{-m}}$
8. $\frac{4}{2^{1-n}}$
9. $\frac{2^{x+1}}{2^x}$
10. $\frac{4^x}{2^{1-x}}$

Exercise 2. Write as a single power of 3:

1. 9×3^p
2. 27^a
3. 3×9^n
4. 27×3^d
5. 9×27^t
6. $\frac{3^y}{3}$
7. $\frac{3}{3^y}$
8. $\frac{9}{27^t}$
9. $\frac{9^a}{3^{1-a}}$
10. $\frac{9^{n+1}}{3^{2n-1}}$

Exercise 3. Consider 3^1, 3^2, 3^3, 3^4, 3^5, ... Look for a pattern and hence find the last digit of 3^{101}.

Exercise 4. What is the last digit of 7^{217}?

Exercise 5. Write the following in simplest form, without brackets:

1. $(-2a)^2$
2. $(-6b^2)^2$
3. $(-2a)^3$
4. $(-3m^2n^2)^3$
5. $(-2ab^4)^4$
6. $\left(\frac{-2a^2}{b^2}\right)^3$

7. $\left(\dfrac{-4a^3}{b}\right)^2$

8. $\left(\dfrac{-3p^2}{q^3}\right)^2$

Exercise 6. Write without negative exponents:

1. ab^{-2}
2. $(ab)^{-2}$
3. $(2ab^{-1})^2$
4. $(3a^{-2}b)^2$
5. $\dfrac{a^2b^{-1}}{c^2}$
6. $\dfrac{a^2b^{-1}}{c^{-2}}$
7. $\dfrac{1}{a^{-3}}$
8. $\dfrac{a^{-2}}{b^{-3}}$
9. $\dfrac{2a^{-1}}{d^2}$
10. $\dfrac{12a}{m^{-3}}$

Remark. Nicomachus discovered an interesting number pattern involving cubes and sums of odd numbers. Nicomachus was born in Roman Syria (now Jerash, Jordan) around 100 AD. He wrote in Greek and was a Pythagorean.

$$1 = 1^3$$

$$3 + 5 = 8 = 2^3$$

$$7 + 9 + 11 = 27 = 3^3$$

$$\vdots$$

Exercise 7. Using Nicomachus' pattern, find the series of odd numbers for:

1. 5^3
2. 7^3
3. 12^3

b - Rational exponents

Definition

$$a^{\frac{1}{n}} = \sqrt[n]{a} \quad \text{where } \sqrt[n]{a} \text{ reads "the } n\text{th root of } a\text{", for } n \in \mathbb{Z}^+.$$

If n is even, we need $a \geq 0$ and $\sqrt[n]{a}$ is the non-negative root.

Property

$$a^{\frac{m}{n}} = \sqrt[n]{a^m} = \left(\sqrt[n]{a}\right)^m \quad \text{for } a > 0,\ n \in \mathbb{Z}^+,\ m \in \mathbb{Z}.$$

The exponent laws above also hold for rational exponents (with $a, b > 0$).

Exercise 8. Write as a single power of 3:

1. $\sqrt[3]{3}$
2. $\dfrac{1}{\sqrt[3]{3}}$
3. $\sqrt[4]{3}$
4. $3\sqrt{3}$
5. $\dfrac{1}{9\sqrt{3}}$

Exercise 9. Without using a calculator, write in simplest rational form:

1. $4^{\frac{3}{2}}$
2. $8^{\frac{5}{3}}$
3. $16^{\frac{3}{4}}$
4. $25^{\frac{3}{2}}$
5. $32^{\frac{2}{5}}$
6. $4^{-\frac{1}{2}}$
7. $9^{-\frac{3}{2}}$
8. $8^{-\frac{4}{3}}$
9. $27^{-\frac{4}{3}}$
10. $125^{-\frac{2}{3}}$

II - Algebra with exponents

a - Expansion

Definition

Usual expansion laws:

$$(a+b)(a-b) = a^2 - b^2$$
$$(a+b)^2 = a^2 + 2ab + b^2$$
$$(a-b)^2 = a^2 - 2ab + b^2$$

Example

Expand and simplify:

$$x^{-\frac{1}{2}}\left(x^{\frac{3}{2}} + 2x^{\frac{1}{2}} - 3x^{-\frac{1}{2}}\right)$$

Example

Expand and simplify:

1. $(2^x+3)(2^x+1)$
2. $(7^x+7^{-x})^2$

Exercise 10. Expand and simplify:

1. $(2^x-1)(2^x+3)$
2. $(3^x+2)(3^x+5)$
3. $(5^x-2)(5^x-4)$
4. $(2^x+3)^2$
5. $(3^x-1)^2$
6. $(4^x+7)^2$
7. $(x^{\frac{1}{2}}+2)(x^{\frac{1}{2}}-2)$
8. $(2^x+3)(2^x-3)$
9. $(x^{\frac{1}{2}}+x^{-\frac{1}{2}})(x^{\frac{1}{2}}-x^{-\frac{1}{2}})$
10. $\left(x+\frac{2}{x}\right)^2$
11. $(7^x-7^{-x})^2$
12. $(5-2^{-x})^2$

b - Factorisation

Example

Factorise:

1. $2^{n+3} + 2^n$
2. $2^{n+3} + 8$
3. $2^{3n} + 2^{2n}$

Example

Factorise:

1. $4^x - 9$
2. $9^x + 4(3^x) + 4$

Exercise 11. Factorise:

1. $9^x - 4$
2. $4^x - 25$
3. $16 - 9^x$
4. $25 - 4^x$
5. $9^x - 4^x$
6. $4^x + 6(2^x) + 9$
7. $9^x + 10(3^x) + 25$
8. $4^x - 14(2^x) + 49$
9. $25^x - 4(5^x) + 4$

Exercise 12. Factorise:

1. $4^x + 9(2^x) + 18$
2. $4^x - 2^x - 20$
3. $9^x + 9(3^x) + 14$
4. $9^x + 4(3^x) - 5$
5. $25^x + 5^x - 2$
6. $49^x - 7^{x+1} + 12$

Example

Simplify:

1. $\dfrac{3^n + 6^n}{3^n}$
2. $\dfrac{2^{m+2} - 2^m}{2^m}$
3. $\dfrac{2^{m+3} + 2^m}{9}$

Exercise 13. Simplify:

1. $\dfrac{6^m + 2^m}{2^m}$
2. $\dfrac{2^n + 12^n}{2^n}$
3. $\dfrac{8^n + 4^n}{2^n}$
4. $\dfrac{12^x - 3^x}{3^x}$
5. $\dfrac{6^n + 12^n}{1 + 2^n}$
6. $\dfrac{5^{n+1} - 5^n}{4}$
7. $\dfrac{5^{n+1} - 5^n}{5^n}$
8. $\dfrac{4^n - 2^n}{2^n}$
9. $\dfrac{2^n - 2^{n-1}}{2^n}$

Exercise 14. Simplify:

1. $2^n(n+1) + 2^n(n-1)$
2. $3^n\left(\dfrac{n-1}{6}\right) - 3^n\left(\dfrac{n+1}{6}\right)$

III - Exponential equations

Property

If the base numbers are the same, we can **equate exponents**.

If $a^x = a^k$ then $x = k$ $(a > 0,\ a \neq 1)$.

Example

Solve for x:

1. $2^x = 16$
2. $3^{x+2} = \dfrac{1}{27}$

Example

Solve for x:

1. $4^x = 8$
2. $9^{x-2} = \dfrac{1}{3}$

Exercise 15. Solve for x, if possible:

1. $4^{2x+1} = 8^{1-x}$
2. $9^{2-x} = \left(\dfrac{1}{3}\right)^{2x+1}$
3. $2^x \cdot 8^{1-x} = \dfrac{1}{4}$

Exercise 16. Solve for x:

1. $3 \cdot 2^x = 24$
2. $7 \cdot 2^x = 56$
3. $3 \cdot 2^{x+1} = 24$
4. $12 \cdot 3^{-x} = \dfrac{4}{3}$
5. $4 \cdot \left(\dfrac{1}{3}\right)^x = 36$
6. $5 \cdot \left(\dfrac{1}{2}\right)^x = 20$

Example

Solve for x: $4^x + 2^x - 20 = 0$

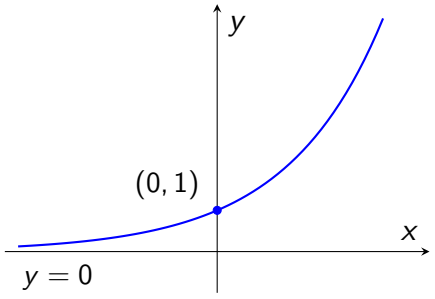
Exercise 17. Solve for x:

1. $4^x - 6(2^x) + 8 = 0$
2. $4^x - 2^x - 2 = 0$
3. $9^x - 12(3^x) + 27 = 0$
4. $9^x = 3^x + 6$
5. $25^x - 23(5^x) - 50 = 0$
6. $49^x + 1 = 2(7^x)$

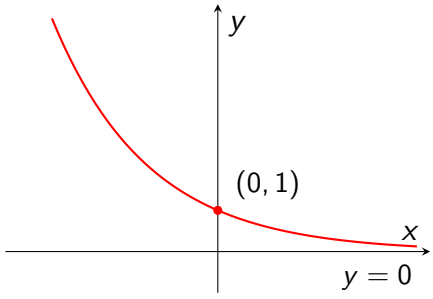
IV - Exponential functions

Definition

The simplest general **exponential function** has the form $y = b^x$ **where** $b > 0$, $b \neq 1$.



$y = b^x,\ b > 1$



$y = b^x,\ 0 < b < 1$

Investigation

1. **Explore the family of curves of the form $y = b^x$ where $b > 0$.**
   For example, consider $y = 2^x$, $y = 3^x$, $y = 10^x$, and $y = (1.3)^x$.
   (a) What effect does changing b have on the shape of the graph?
   (b) What is the y-intercept of each graph?
   (c) What is the horizontal asymptote of each graph?
2. **Explore the family of curves of the form $y = 2^x + d$ where d is a constant.**
   For example, consider $y = 2^x$, $y = 2^x + 1$, and $y = 2^x - 2$.
   (a) What effect does changing d have on the position of the graph?
   (b) What effect does changing d have on the shape of the graph?

(c) What is the horizontal asymptote of each graph?

(d) What is the horizontal asymptote of $y = 2^x + d$?

(e) To graph $y = 2^x + d$ from $y = 2^x$, what transformation is used?

3. **Explore the family of curves of the form $y = 2^{x-c}$.**
   For example, consider $y = 2^x$, $y = 2^{x-1}$, $y = 2^{x+2}$, and $y = 2^{x-3}$.

   (a) What effect does changing c have on the position of the graph?

   (b) What effect does changing c have on the shape of the graph?

   (c) What is the horizontal asymptote of each graph?

   (d) To graph $y = 2^{x-c}$ from $y = 2^x$, what transformation is used?

4. **Explore the relationship between $y = b^x$ and $y = b^{-x}$ where $b > 0$.**
   For example, consider $y = 2^x$ and $y = 2^{-x}$.

   (a) What is the y-intercept of each graph?

   (b) What is the horizontal asymptote of each graph?

   (c) What transformation moves $y = 2^x$ to $y = 2^{-x}$?

5. **Explore the family of curves of the form $y = a \times 2^x$ where a is a constant.**

   (a) Consider functions where $a > 0$, such as $y = 2^x$, $y = 3 \times 2^x$, and $y = \frac{1}{2} \times 2^x$.
   Comment on the effect on the graph.

   (b) Consider functions where $a < 0$, such as $y = -2^x$, $y = -3 \times 2^x$, and $y = -\frac{1}{2} \times 2^x$.
   Comment on the effect on the graph.

   (c) What is the horizontal asymptote of each graph? Explain your answer.

Property

For the general exponential function $y = a \times b^{x-c} + d$ where $b > 0,\ b \neq 1,\ a \neq 0$:

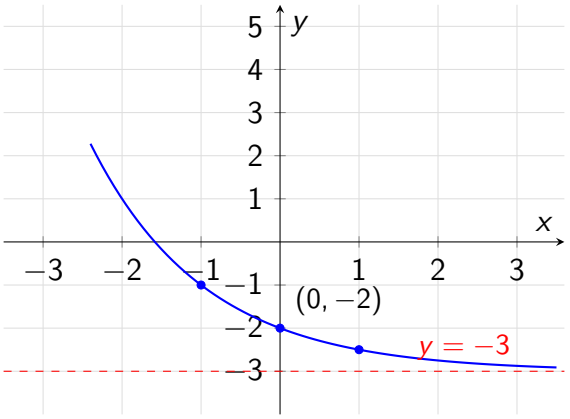
- b controls how steeply the graph increases or decreases
- c controls horizontal translation
- d controls vertical translation
- the equation of the horizontal asymptote is $y = d$
- if $a > 0,\ b > 1$, the function is increasing.
- if $a > 0,\ 0 < b < 1$, the function is decreasing.
- if $a < 0,\ b > 1$, the function is decreasing.
- if $a < 0,\ 0 < b < 1$, the function is increasing.

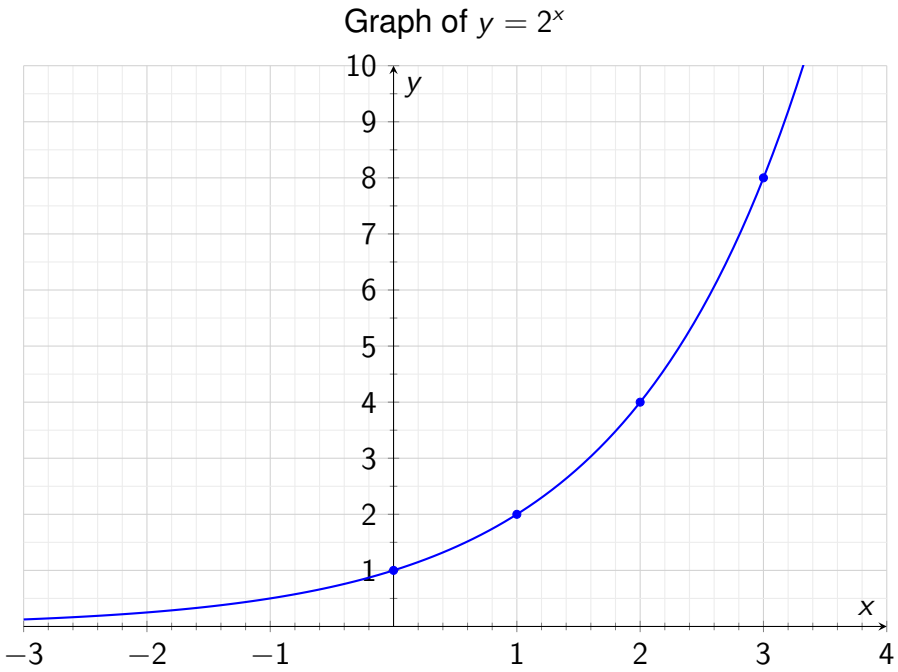
We can now sketch reasonably accurate graphs of exponential functions using:

- the horizontal asymptote
- the y-intercept
- two other points.

Example

Sketch the graph of $y = 2^{-x} - 3$. Hence state the domain and range of $f(x) = 2^{-x} - 3$.





Exercise 18. Use the graph of $y = 2^x$ above to estimate the value of:

1. $2^{\frac{1}{2}}$ or $\sqrt{2}$
2. $2^{0.8}$
3. $2^{1.5}$

Exercise 19. Use the graph of $y = 2^x$ above to estimate the solution to:

1. $2^x = 3$
2. $2^x = 0.6$

Exercise 20. Use the graph of $y = 2^x$ to explain why $2^x = 0$ has no solutions.

Exercise 21. Draw freehand sketches of the following pairs of graphs:

1. $y = 3^x$ and $y = 3^{-x}$
2. $y = 3^x$ and $y = 3^x + 1$
3. $y = 3^x$ and $y = -3^x$
4. $y = 3^x$ and $y = 3^{x-1}$

Exercise 22. For each of the functions below:

1. sketch the graph of the function

2. state the domain and range
3. use your calculator to find the value of y when $x = \sqrt{2}$
4. discuss the behaviour of y as $x \to \pm\infty$
5. determine the horizontal asymptotes

(i) $y = 2^x + 1$

(ii) $y = 2 - 2^x$

(iii) $y = 2^{-x} + 3$

(iv) $y = 3 - 2^{-x}$

V - Growth and decay

Populations of animals, people, and bacteria usually *grow* in an exponential way.
Radioactive substances, and items that depreciate in value, usually *decay* exponentially.

Example

An entomologist monitoring a grasshopper plague notices that the area affected by the grasshoppers is given by $A_n = 1000 \times 2^{0.2n}$ hectares, where n is the number of weeks after the initial observation.

1. Find the original affected area.
2. Find the affected area after:
   (a) 5 weeks
   (b) 10 weeks
   (c) 12 weeks.
3. Draw the graph of A_n against n.

Example

When a diesel-electric generator is switched off, the current dies away according to the formula $I(t) = 24 \times (0.25)^t$ amps, where t is the time in seconds after the power is cut.

1. Find $I(t)$ when $t = 0, 1, 2$ and 3.
2. What current flowed in the generator at the instant when it was switched off?
3. Plot the graph of $I(t)$ for $t \geq 0$ using the information above.
4. Use your graph or technology to find how long it takes for the current to reach 4 amps.

Example

The weight of radioactive material remaining after t years is given by $W_t = W_0 \times 2^{-0.001t}$ grams.

1. Find the original weight.
2. Find the percentage remaining after 200 years.

Exercise 23. A species of bear is introduced to a large island off Alaska where previously there were no bears. 6 pairs of bears were introduced in 1998. It is expected that the population will increase according to $B_t = B_0 \times 2^{0.18t}$ where t is the time since the introduction.

1. Find B_0.
2. Find the expected bear population in 2018.
3. Find the expected percentage increase from 2008 to 2018.

Exercise 24. The speed V_t of a chemical reaction is given by $V_t = V_0 \times 2^{0.05t}$ where t is the temperature in $^\circ$C.

1. Find the reaction speed at:
   (a) $0°C$
   (b) $20°C$.
2. Find the percentage increase in reaction speed at $20°C$ compared with $0°C$.
3. Find $\left(\frac{V_{50} - V_{20}}{V_{20}}\right) \times 100\%$ and explain what this calculation means.

Exercise 25. The weight W_t grams of radioactive substance remaining after t years is given by $W_t = 1000 \times 2^{-0.03t}$ grams.

1. Find the initial weight of the radioactive substance.
2. Find the weight remaining after:
   (a) 10 years
   (b) 100 years
   (c) 1000 years.
3. Graph W_t against t using (a) and (b) only.
4. Use your graph or graphics calculator to find the time when 10 grams of the substance remains.
5. Write an expression for the amount of substance that has decayed after t years.

Exercise 26. The weight W_t of a radioactive uranium-235 sample remaining after t years is given by the formula $W_t = W_0 \times 2^{-0.0002t}$ grams, $t \geq 0$. Find:

1. the original weight
2. the percentage weight loss after 1000 years
3. the time required until $\frac{1}{512}$ of the sample remains.

VI - Natural exponential

Investigation

A formula for calculating the amount to which an investment grows is

$$u_n = u_0(1+i)^n$$

where:

- u_n is the final amount,
- u_0 is the initial amount,
- i is the interest rate per compounding period,
- n is the number of periods or number of times the interest is compounded.

We will investigate the final value of an investment for various values of n, and allow n to get extremely large.

1. Suppose \$1000 is invested for one year at a fixed rate of 6% per annum. Use your calculator to find the final amount or *maturing value* if the interest is paid:
   (a) annually $(n = 1,\ i = 6\% = 0.06)$
   (b) quarterly $(n = 4,\ i = \dfrac{6\%}{4} = 0.015)$
   (c) monthly
   (d) daily
   (e) by the second
   (f) by the millisecond.
2. Comment on your answers from 1.
3. If r is the percentage rate per year (as a decimal), t is the number of years, and N is the number of interest payments per year, then

   $$i = \frac{r}{N} \qquad \text{and} \qquad n = Nt.$$

   The growth formula becomes

   $$u_n = u_0\left(1+\frac{r}{N}\right)^{Nt}.$$

   If we let $a = \dfrac{N}{r}$, show that

   $$u_n = u_0\left[\left(1+\frac{1}{a}\right)^{a}\right]^{rt}.$$

4. For continuous compound growth, the number of interest payments per year N gets very large.

   (a) Explain why a gets very large as N gets very large.

   (b) Copy and complete the table, giving your answers as accurately as technology permits.

a	$\left(1+\frac{1}{a}\right)^a$
10	
100	
1000	
10 000	
100 000	
1 000 000	
10 000 000	

5. You should have found that for very large values of a,

$$\left(1+\frac{1}{a}\right)^a \approx 2.718281828459\ldots$$

   Use the $\boxed{e^x}$ key of your calculator to find the value of e^1. What do you notice?

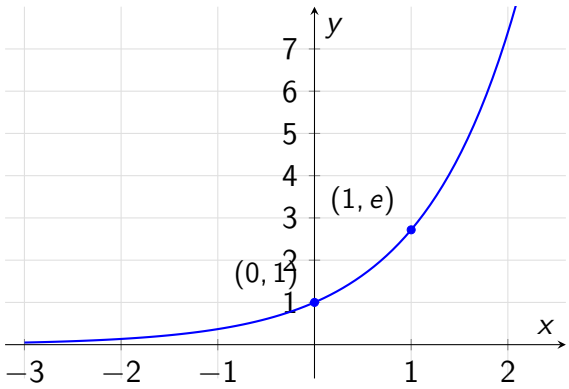
6. For continuous growth,

$$u_n = u_0 e^{rt}$$

   where u_0 is the initial amount, r is the annual percentage rate (as a decimal), and t is the number of years.

   Use this formula to find the final value if \$1000 is invested for 4 years at a fixed rate of 6% per annum, where the interest is calculated continuously.

Graph of $y = e^x$



Exercise 27. Sketch, on the same set of axes, the graphs of $y = e^x$ and $y = e^{-x}$.

What is the geometric connection between these two graphs?

Exercise 28. For the general exponential function $y = ae^{kx}$, what is the y-intercept?

Exercise 29. Consider $y = 2e^x$.

1. Explain why y can never be < 0.
2. Find y if:
   (a) $x = -20$
   (b) $x = 20$.

Exercise 30. Find, to 3 significant figures, the value of:

1. e^2
2. e^3
3. $e^{0.7}$
4. $\sqrt{e}$
5. e^{-1}

Exercise 31. Write the following as powers of e:

1. $\sqrt{e}$
2. $\dfrac{1}{\sqrt{e}}$
3. $\dfrac{1}{e^2}$
4. $e\sqrt{e}$

Exercise 32. Simplify:

1. $(e^{0.36})^{\frac{t}{2}}$
2. $(e^{0.064})^{\frac{t}{16}}$
3. $(e^{-0.04})^{\frac{t}{8}}$
4. $(e^{-0.836})^{\frac{t}{5}}$

Exercise 33. Find, to five significant figures, the values of:

1. $e^{2.31}$
2. $e^{-2.31}$
3. $e^{4.829}$
4. $e^{-4.829}$
5. $50e^{-0.1764}$
6. $80e^{-0.6342}$
7. $1000e^{1.2642}$
8. $0.25e^{-3.6742}$

Exercise 34. Solve for x:

1. $e^x = \sqrt{e}$
2. $e^{\frac{1}{2}x} = \dfrac{1}{e^2}$

Exercise 35. The current flowing in an electrical circuit t seconds after it is switched off is given by $I(t) = 75e^{-0.15t}$ amps.

1. What current is still flowing in the circuit after:
   (a) 1 second
   (b) 10 seconds?
2. Use your graphics calculator to sketch $I(t) = 75e^{-0.15t}$ and $I = 1$.
3. Hence find how long it will take for the current to fall to 1 amp.

Exercise 36. Consider the function $f(x) = e^x$.

1. On the same set of axes, sketch $y = f(x)$, $y = x$, and $y = f^{-1}(x)$.
2. State the domain and range of f^{-1}.